

BREATHEBOX · DESIGN DECISION RECORD

BreatheBox recommendations accepted

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DOCUMENT CONTROL

Document ID	BBX-DDR-002	Version	0.2
Status	Draft	Date	2026-09-26
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breathebox		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-26	Amish Chadha	Budget set to \$255 to cover the priced BOM: decided by Amish, 2026-09-26 (O3 closed)
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Amish accepted every recommendation in docs/REVIEW.md and BBX-DDR-001. Items without a recommendation stay proposed, awaiting Amish, except the budget gap (O3), decided by Amish on 2026-09-26.

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item in docs/REVIEW.md and BBX-DDR-001 that was awaiting Amish or adopted for TRL 3 pending his review, and that carried a recommendation, is now decided: go with recommendation. Where a recommendation offered several options, the recommended option is the decision. Items with no recommendation are not decided here. TRL 4 remains on hold by Amish's instruction, so decided items that need building, testing or purchasing are recorded as decided but on hold.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item (source)	Decision	What changed in the repo
1	Budget (TRL 2 item 1; DDR-001 A1)	Raise <code>budget_usd</code> from \$220 to \$250; keep the bought core and the CO2 sensor	<code>project.yaml</code> <code>budget_usd</code> 220 to 250; README budget line; BBX-REQ-001 R15 target \$250; BBX-CAL-001 section 9 and <code>results.csv</code> ; BBX-PRB-001 constraints; <code>bom/bom-notes.md</code> ; blueprint key figures. R15 is still not met (\$255, \$5 over)
2	Window type (TRL 2 item 2; A2)	Vertical sliding sash first; casement and tilt-and-turn later	Wording only (BBX-PRB-001, BBX-REQ-001 R8, BBX-PRC-001); design already sash-first
3	Core type (TRL 2 item 3; A3)	Bought polymer counterflow plate core	Wording only (BBX-PRC-001); already in BOM line 2
4	Frost strategy (TRL 2 item 4; A4)	Slow the supply fan, with the combustion-appliance warning	Wording only; already in BBX-CAL-001 section 6 and the safety notes
5	Nominal airflow (TRL 2 item 5; A5)	50 m ³ /h with CO2-driven boost to 70 m ³ /h	Wording only; already the design case
6	Controller (TRL 2 item 6; A6)	ESP32-C3 class module with an SCD41 class sensor, checked on CalRig	Wording only. The CalRig sensor check is TRL 4 work: decided, on hold
7	Noise, R6 (TRL 3 item 2; DDR-001 O2)	Option (a) quiet night mode near 30 to 32 m ³ /h, plus option (b) larger, slower fans or a lined silencer checked against a real fan datasheet before any change to R6	Firmware rule added: night mode holds both streams at 32 m ³ /h and suppresses the CO2 boost. BBX-CAL-001 v0.2 (<code>sizing.py</code>): 30.1 dB(A) clean, 32.0 dB(A) loaded; CO2 1,295 ppm overnight, 712 ppm 24 h mean. BBX-PRC-001, BBX-REQ-001 R4 and R6 status, BBX-DWG-001 note (Rev P1 to P2), blueprint key figures, README. R6 target unchanged and still not met at 50 m ³ /h. Option (b) needs a chosen fan and its datasheet: decided, on hold with TRL 4
8	Larger blowers, about 100 m ³ /h free air and 400 Pa (TRL 3 item 4)	Keep	Already in BOM lines 3 and 4 and BBX-CAL-001; wording updated in BBX-PRC-001
9	2 A time-delay input fuse (TRL 3 item 4)	Keep	Already in BOM line 13; wording updated in BBX-PRC-001
10	Firmware fan speed cap at the 80 m ³ /h need (TRL 3 item 4)	Adopt as a firmware rule	BBX-PRC-001 (How it works, step 6) and BBX-CAL-001 section 4 now state it as a rule. No firmware is written (TRL 4, on hold)
11	Install restriction for rooms whose door seals airtight (TRL 3 item 4)	Keep	Already in the safety notes of BBX-PRC-001 and BBX-CAL-001

No recommendation changed the pitch or problem wording, so `project.yaml` and `README.md` keep them.

Items still open

Table 2. Items still proposed, awaiting Amish (no recommendation was made).

#	Item	Status
O1	First co-design partner and region (for example a tenants' group or social landlord in the UK or Canada)	Proposed, awaiting Amish
O3	Budget gap: priced BOM \$255 against the new \$250 (accept about \$255, drop to smaller blowers and lose the 70 m³/h boost, or find \$5 elsewhere)	Decided by Amish, 2026-09-26: budget set to \$255 (see below)
O4	Fan position: both fans in the room-end plenum (exhaust side of the core at higher pressure, so any core leakage reaches the supply) versus the exhaust fan at the outdoor end	Proposed, awaiting Amish. No change made

Budget approved, 2026-09-26

On 2026-09-26 Amish wrote: "i approve all the budget items."

- Budget set to \$255 to cover the priced BOM: decided by Amish, 2026-09-26. This closes O3. `project.yaml` `budget_usd` 250 to 255; BBX-REQ-001 R15 target \$255, status Not met to Met; BBX-CAL-001 v0.3 (`sizing.py` and `results.csv` rerun); BBX-PRB-001, BBX-PRC-001, `README`, `bom/bom-notes.md` and the blueprint key figures updated. The BOM and geometry are unchanged.

Consequences

- Requirement status (BBX-CAL-001 v0.3): met 11, not met 1 (R6 at 50 m³/h), not verifiable at TRL 3 3 (R9, R13, R14). The night mode narrows the R6 gap in use but not as R6 is written. R15 is met since the 2026-09-26 budget approval.
- The geometry and BOM are unchanged. The GA drawing moves to Rev P2 for the night mode note only; STEP and STL were re-exported from the unchanged model.
- Cross-repo: the SCD41 class sensor check on CalRig is decided and on hold with TRL 4. No other repo is affected, and none was edited.
- `trl: 3` and `trl_target: 3` are unchanged.